LO1 ASSIGNMENT

Question 1: AI Fundamentals and Project Scoping (15 Marks)

This question tests your understanding of core AI concepts and the initial project planning/documentation phase.

a) AI Foundations: (5 Marks)

i. Briefly state one key difference between Artificial Intelligence (AI) and Machine Learning (ML). (2 Marks)

ii. Why is the historical overview of AI important for new developers? (1 Mark)

iii. Explain the importance of integrating AI into electronic devices today, providing a specific example of an industry that benefits. (2 Marks)

b) Project Documentation: (10 Marks)

You are leading the project to develop a Smart Surveillance System that uses a camera to distinguish between humans and pets.

i. In reference to this problem, analyze and detail **one specific Social factor, one Economical factor**, and **one Technical challenge that the AI solution is designed to address**. (3 Marks)

ii. List three (3) critical sources of information you would consult to gather technical requirements and background knowledge for this project. (3 Marks)

iii. Briefly describe the core function of the AI documentation process and list two (2) key elements that must be recorded at this initial stage. (4 Marks)

Question 2: Data Acquisition and Pre-processing (15 Marks)

This question focuses on the practical steps of gathering and cleaning the data before training.

a) Data Collection: (5 Marks)

i. The Smart Surveillance System needs images. What is the appropriate AI data category for this type of project? (1 Mark)

ii. Describe two (2) distinct AI-Data collection techniques you would use to acquire the necessary images of humans and pets for your initial dataset. (4 Marks)

b) Data Pre-processing: (10 Marks)

After collecting the images, you must perform Data Pre-processing. Explain the purpose of the three main operations and provide a specific example of what each one would entail when applied to your image dataset.

i. Cleaning: (3 Marks)

ii. Filtering: (3 Marks)

iii. Data Transformation: (4 Marks)

Question 3: File Preparation and Tooling (10 Marks)

This question covers the final steps of preparing the file and importing it into the development environment.

a) Data File Creation: (5 Marks)

i. Suggest an appropriate Data format (e.g., JPEG, PNG, CSV) for the final dataset file that organizes your cleaned and labeled images. Justify your choice based on the needs of an image classification project. (2 Marks)

ii. Briefly outline the File creation process—the logical steps you take to structure your pre-processed images and their labels into a file or directory structure ready for import. (3 Marks)

b) Dataset Import: (5 Marks)

i. Name one specific Environment software (IDE) (e.g., from the list: Anaconda, Visual Studio Code) and one hypothetical DATASET Importing technique (e.g., a function call) you would use to load the data into your program. (2 Marks)

ii. When preparing for training, you will divide your dataset. What are the two (2) main Types of dataset that are created from the original collected data? (1 Mark)

iii. Explain the role of the chosen Development tools (like Anaconda or VS Code) in performing the data import successfully. (2 Marks)